

Jonathan von Rad

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in LinkedIn 🐙 GitHub 📄 Google Scholar

Research Area

- NLP, LLMs, Multilinguality, AI Safety, Reinforcement Learning, Model Efficiency, Interpretability
- PhD offer from the University of Cambridge for multilingual NLP research at the Language Technology Lab

Education

University College London

09/2025 – 09/2026

M.Sc. Machine Learning, Grade: Distinction (75%)

London, UK

- Coursework: Reinforcement Learning (DeepMind), Unsupervised Learning, Statistical NLP, Supervised Learning, Machine Vision, Openendedness and General Intelligence, Applied ML
- Thesis: Disentangling Tokenizer Quality from Script Disparity in Multilingual LLM Pretraining

University of Tübingen

09/2021 – 08/2025

B.Sc. Computer Science, First Class Honours (1.4/5.0; 1.0 being the best)

Tübingen, Germany

- Ranked in the top 5% of undergraduate computer science students
- Coursework: Linear Algebra, Algorithms & Data Structures, Statistics, Artificial Intelligence
- Thesis: Analyzing Pruning, Quantization and Distillation for LLM Compression (Grade 1.0, GPA 4.0/4.0)

Peking University 北京大学

09/2024 – 06/2025

Exchange Student: Studied Mandarin for one year (B2, HSK-5)

Beijing, China

- Research Assistant in LLM Alignment at PKU Alignment and Interaction Lab

University of Tunis El Manar جامعة تونس المنار

03/2023 – 12/2023

Visiting Student: Studied Arabic for one year (B2)

Tunis, Tunisia

Schelztor Gymnasium Esslingen

09/2016 – 07/2021

Abitur with majors in Maths, Physics and PE, Grade: 1.3 (GPA 4.0/4.0)

Esslingen, Germany

Publications

1. **Jonathan von Rad**, Yong Cao, Andreas Geiger. “UniComp: A Unified Evaluation Framework for LLM Compression via Pruning, Quantization and Distillation”. **EMNLP Main**, 2026.
2. **Jonathan von Rad**, Louis Arts, George Burgess, Eleftheria Kolokytha, Harry O'Donnell, Ektor Oikonomidis Doumpas, Eduardo Sanchez, Yao Lu, Pontus Stenetorp. “PolyFact: Comparing Consistency-Driven Post-Training for Cross-Lingual Factual Recall”. Under review at EACL 2027. [arXiv:2606.06586](https://arxiv.org/abs/2606.06586).
3. **Jonathan von Rad**, Florian Seuffert. “Investigating the Effect of Network Pruning on Performance and Interpretability”. Preprint. [arXiv:2409.19727](https://arxiv.org/abs/2409.19727).

Experience

AI Solutions Engineer (Multilingual)

07/2026 – now

Foundations First, Non-Profit

London, UK

- Developing AI-assisted differentiated learning to tackle education inequality at primary schools in Kenya
- Using quantized Tiny-Aya 4B for on-device agentic modeling and multilingual NLU in Swahili

Teaching Assistant for “Practical Programming I and II”

University of Tübingen, Real Virtual Humans & Database Systems Group

09/2022 – 08/2024

Tübingen, Germany

- Taught functional programming in DrRacket and object-oriented programming in Java (~25 students)
- Evaluated and graded weekly assignments and final exams

Web Developer

Coreflow UG

11/2023 – 03/2024

Tübingen, Germany

- Developed modular web applications for robot configuration using SvelteKit and TypeScript

Selected Projects

Multilingual LLM Pretraining

09/2026

M.Sc. Thesis supervised by Dr. Yao Lu and Prof. Pontus Stenetorp

- Pretrained 18 monolingual and bilingual 1B LLMs on FineWeb2-HQ on ISAMBARD-AI supercomputer
- Trained separate tokenizers and analyzed cross-lingual transfer across a fair and starved vocabulary
- Demonstrated that improved tokenization substantially increases cross-lingual representation alignment
- Revealed an alignment-transfer dissociation, where alignment rises but transfer stagnates

PolyFact: Cross-Lingual Factual Consistency

2026

UCL, first author, under review at EACL 2027

[arXiv](#)

- Built a fully parallel QA dataset of 58,807 Wikidata-grounded facts across 12 typologically diverse languages
- Proposed consistency-driven GRPO with cross-lingual reward pooling on OLMo-2-7B and Qwen-2.5-7B
- Achieved the strongest free-form transfer on Qwen (+9.4pp) against SFT, DCO and CM-Align

Emergent Communication in Multi-Agent RL

2026

UCL, Open-Endedness and General Intelligence (group project)

[Code](#) 

- Built a JAX gridworld where a non-acting “Seer” guides two blind agents through a corridor bottleneck
- Trained agents jointly with MAPPO through a 4-bit discrete FSQ channel (16-symbol vocabulary)
- Two-phase curriculum learning raised task success from ~55% to over 90%

Extracurricular Activities

- Reviewer, *Artificial Intelligence Review* (Springer Nature), 2026
- Youth Leader at local church “Gemeindeverbund God in Life”
- Taught English to children in Beijing

Awards

- DAAD Scholarship for master’s abroad at UCL (10% acc. rate; £41,000 total funding) 2025
- Baden-Württemberg Stipendium for a year abroad at Peking University (3,000€ funding) 2024

Skills and Languages

Technical Skills: Python (PyTorch, JAX, NumPy, Pandas), Java, TypeScript, R, Linux, Git, LangGraph

Languages: German (native), English (bilingual), Mandarin, Arabic (advanced), French (intermediate)

Interests: Machine Learning, Triathlon, Chinese and Arabic Language, Basketball, Chess, Piano